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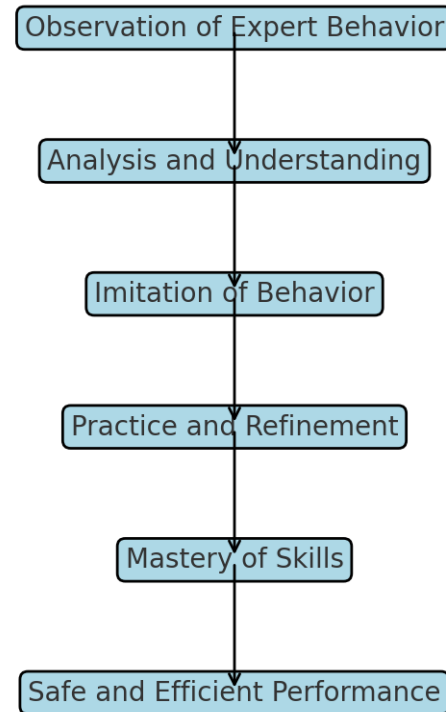


Data-Driven Model-Free Inverse Reinforcement Learning for Discrete-Time Systems

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Efficient Skill Mastery through Imitation and Biomimicry



This approach mirrors nature's design principles (Biomimicry)

Expert Behaviors-Optimal Control Systems



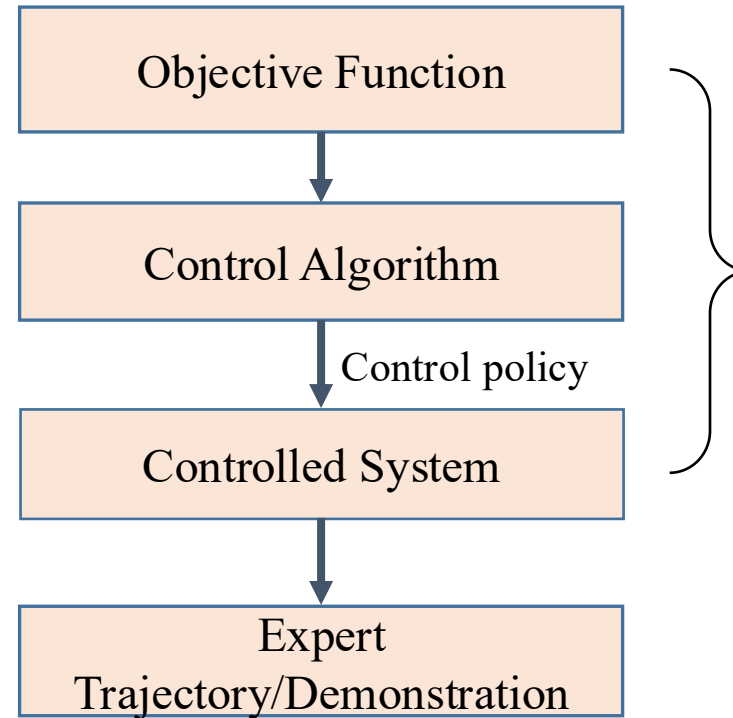
Hamilton



Bellman



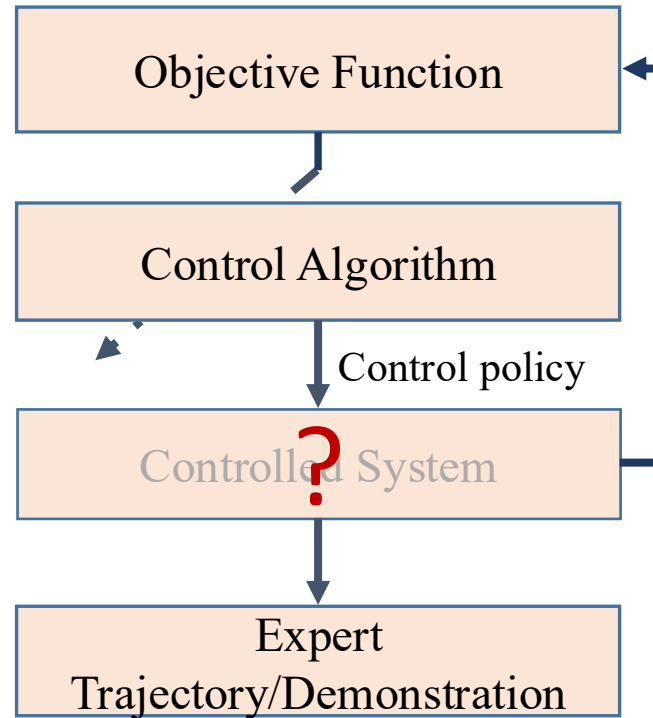
Pontryagin



Optimal control

- Objective function is known

Expert Behaviors-Reinforcement Learning Control



Optimal control

- Objective function is known

Reinforcement learning

- System dynamics is unknown

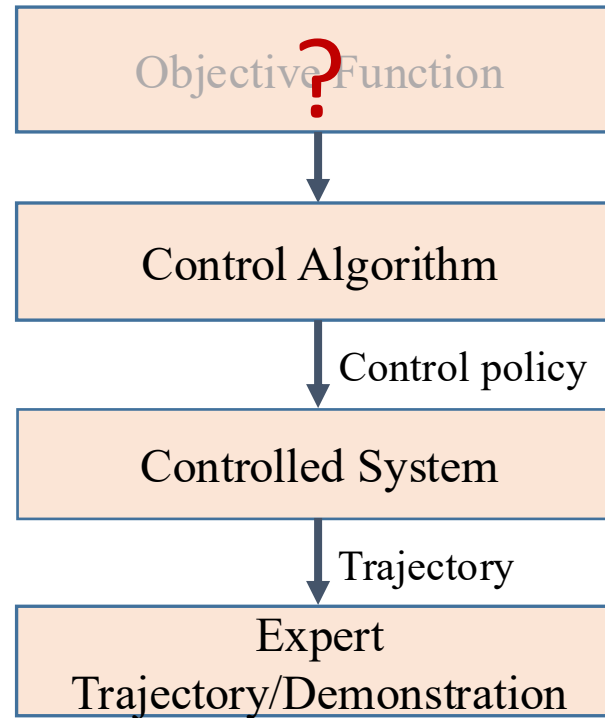
↕ Control systems

Approximate dynamic programming

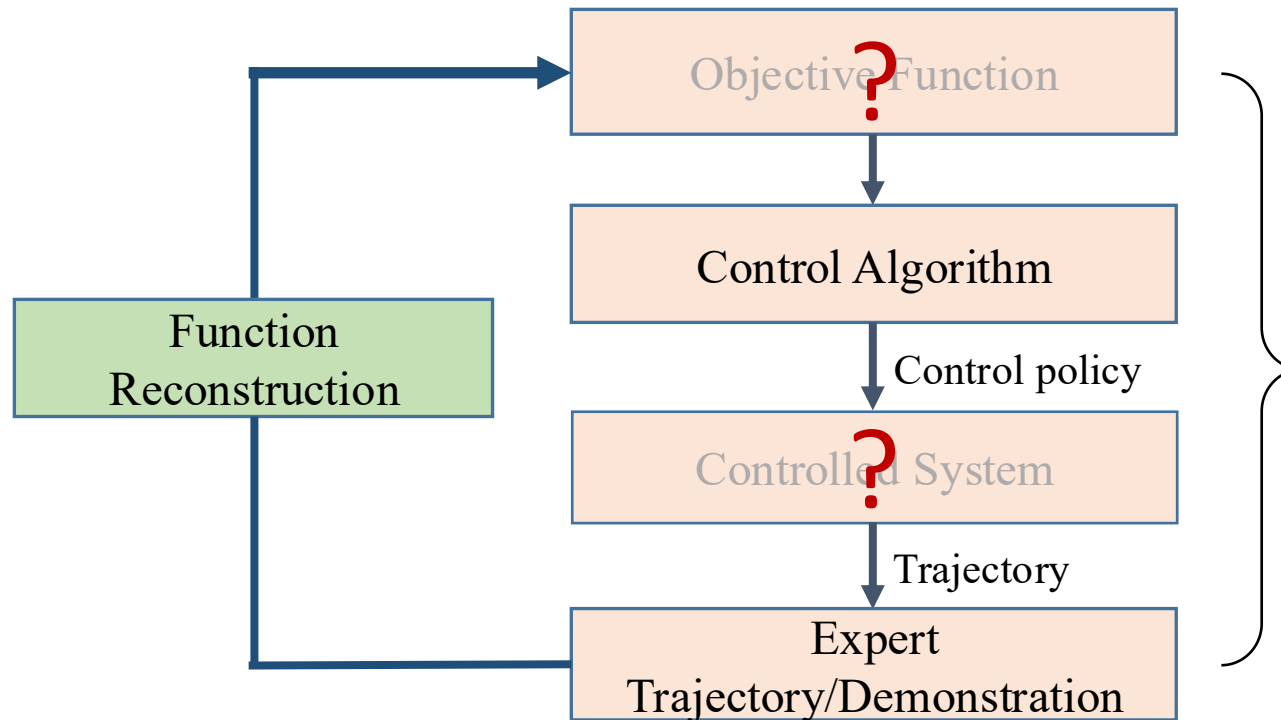
Inverse Problem of Optimal Control

Inverse optimal control

- Objective function is unknown
- Need system dynamics
- Bilevel two-loop learning procedure
---low learning efficiency



Inverse Reinforcement Learning Control



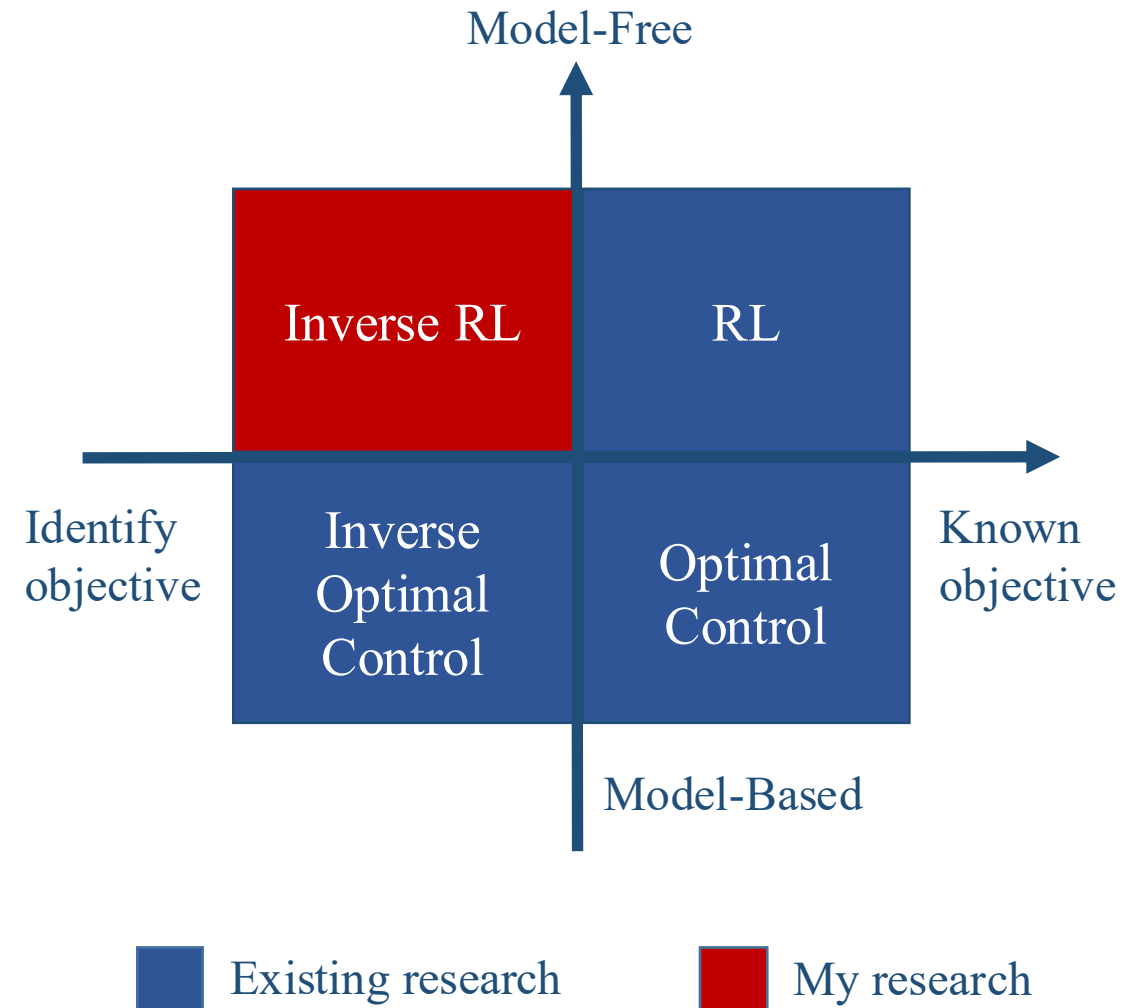
My research:

Data-Driven Model-Free Inverse reinforcement learning for control systems

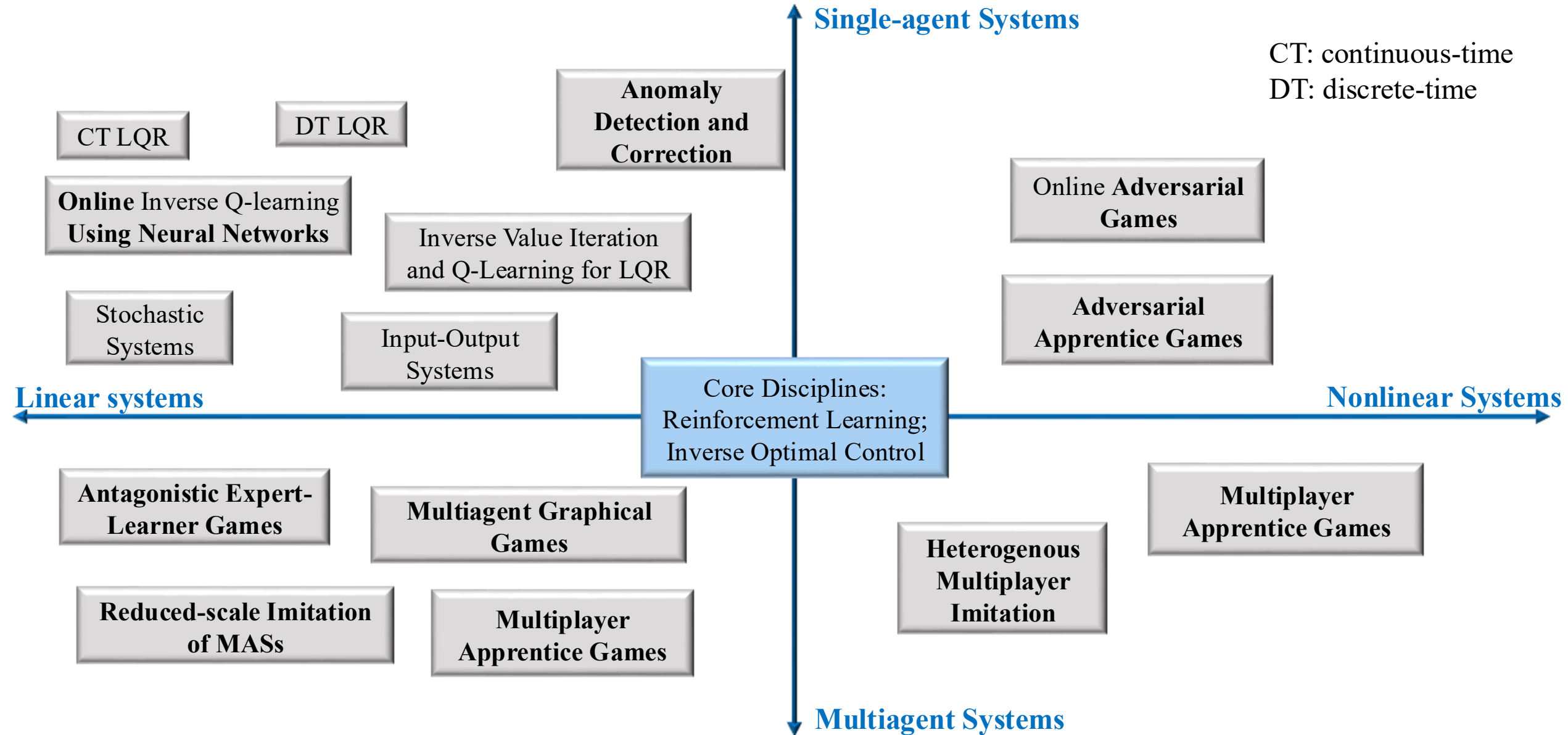
Use only system data to construct function weights and compute control policy to achieve imitation with improved learning efficiency

New Research

■ A new discipline of inverse RL in control theory

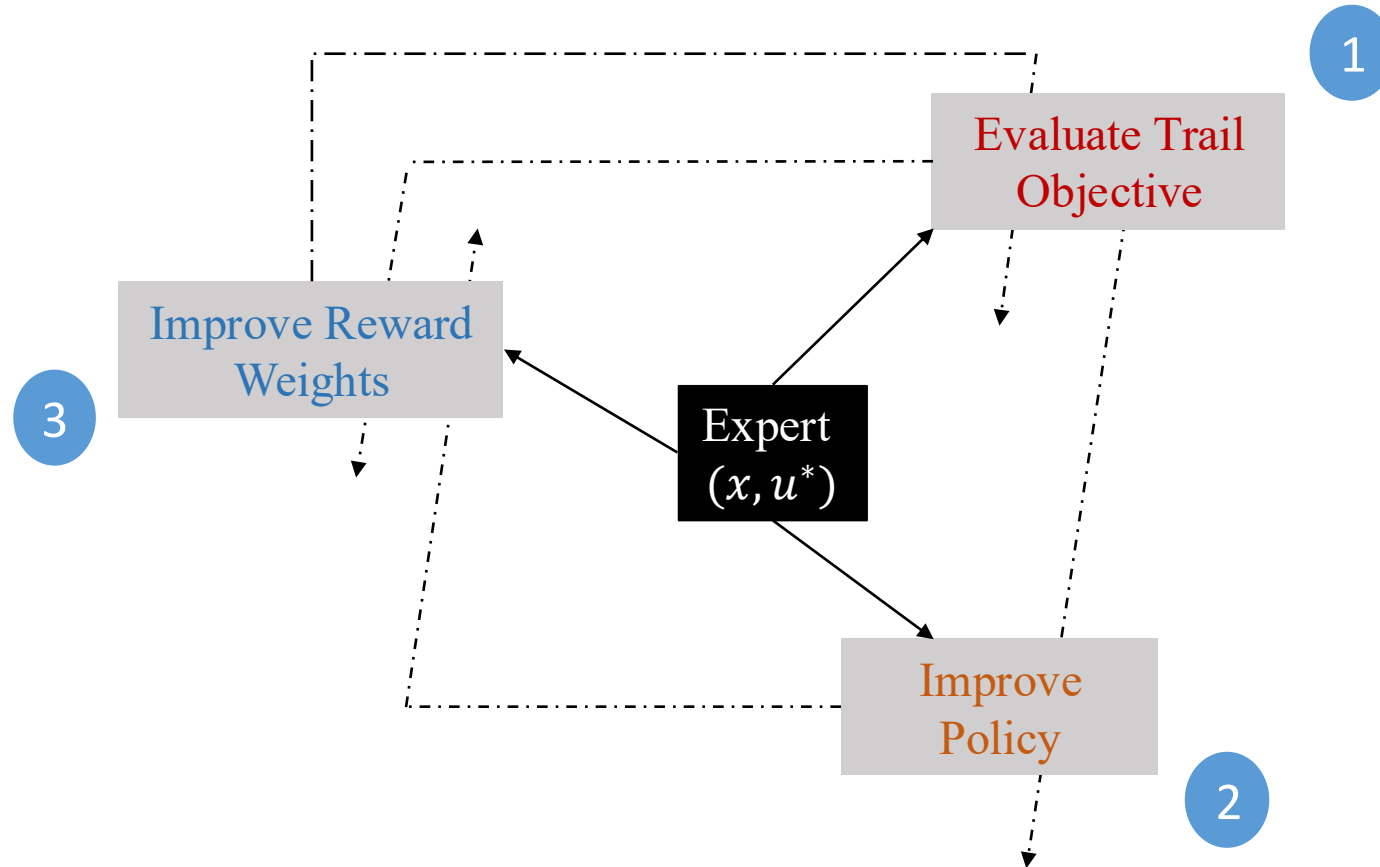


Our Results of Inverse Reinforcement Learning



Inverse RL Diagram

Objective function: Cost function



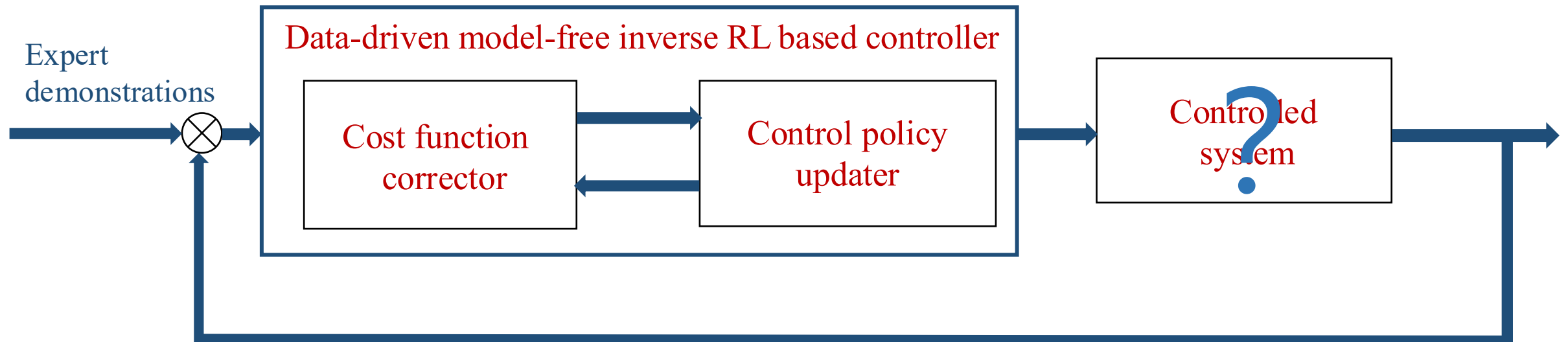
Model-Free Inverse RL for DT State-Feedback Control Systems

■ Controlled System (Learner)

DT linear system with unknown model/dynamics:

$$\begin{cases} x(k+1) = Ax(k) + Bu(k) \\ y(k) = Cx(k) \end{cases}$$

■ Control Problem



Model-Free Inverse RL for DT State-Feedback Control Systems

The expert demonstrations are exhibited by an Expert system with the same dynamics

$$\begin{cases} x_T(k+1) = Ax_T(k) + Bu_T(k) \\ y_T(k) = Cx_T(k) \end{cases}$$

where the control input u_T optimally corresponds to the **underlying expert cost function**

$$V_T(x_T(k), u_T(k)) = \sum_{j=k}^{\infty} (x_T(j)^T Q_T x_T(j) + u_T(j)^T R_T u_T(j))$$

which is given by

$$u_T = -K_T x_T = -(R_T + B^T P_T B)^{-1} B^T P_T A x_T$$

and satisfies the DT algebraic Riccati equation

$$Q_T - P_T + A^T P_T A - A^T P_T B (R_T + B^T P_T B)^{-1} B^T P_T A = 0.$$

or Lyapunov equation

$$Q_T + (A - BK_T)^T P_T (A - BK_T) + K_T^T P_T K_T - P_T = 0.$$

Only measurement data of expert's x_T, u_T, y_T are available

Model-Free Inverse RL for DT State-Feedback Control Systems

■ Control Problem

Using expert's data, find a group of $Q \geq 0$, $R > 0$ and P, K for learner to minimize the following index without knowing system dynamics A, B

$$J^* = \min_u \sum_{t=0}^{\infty} \|x(k) - x_T(k)\| + \|u(k) - u_T(k)\| \quad \longrightarrow \quad J^* = \min_{Q,R} \sum_{t=0}^{\infty} \|x(k) - x_T(k)\| + \|u(k) - u_T(k)\|$$

$$\text{s.t.} \quad \begin{aligned} x(k+1) &= Ax(k) + Bu(k) \\ y(k) &= Cx(k) \end{aligned}$$

$$\text{s.t.} \quad x(k+1) = Ax(k) + Bu(k), \quad y(k) = Cx(k)$$

$$\text{Optimality condition} \quad \begin{cases} Q - P + A^T P A - A^T P B (R + B^T P B)^{-1} B^T P A = 0 \\ u(k) = -Kx(k) = -(R + B^T R B)^{-1} B^T P A x(k) - R^{-1} B^T P x \end{cases}$$

1

2

■ Controller Design

Goal solution condition

Theorem: If there exists a group of Q and R satisfying both (1) and the Lyapunov equation

$$Q + (A - BK_T)^T P (A - BK_T) + K_T^T R K_T - P = 0$$

3

then, the control gain K in (2) equals expert's K_T and minimizes J .

Model-Free Inverse RL for DT State-Feedback Control Systems

■ Controller Design

Model-based iteration framework

1. **Initialization:** select $R > 0$, an initial $Q^0 = 0$, and a threshold $\varepsilon > 0$, set $i = 0$, $\alpha \in (0, 1]$

2: **Control policy evaluation and update:** update P^{i+1} by

$$Q^i - P^{i+1} + (A - BK_T)^T P^{i+1} (A - BK_T) + K_T^T R K_T = 0.$$

and control input u^{i+1} by

$$u^{i+1} = -K^{i+1}x = -(R + B^T P^{i+1} B)^{-1} B^T P^{i+1} A x$$

} **Modified RL policy iteration**

3. **Cost function weight correction:** update Q^{i+1} by

$$Q^{i+1} - (1 - \alpha)Q^i + \alpha(-P^{i+1} + A^T P^{i+1} A - A^T P^{i+1} B (R + B^T P^{i+1} B)^{-1} B^T P^{i+1} A) = 0 \quad \text{Inverse optimal control}$$

4. Set $i = i + 1$, repeat 1-3 until $\|Q^{i+1} - Q^i\| \leq \varepsilon$.

Model-Free Inverse RL for DT State-Feedback Control Systems

■ Control Algorithm

Data-driven model-free inverse reinforcement Q-learning algorithm

1. Initialization: Set $i = 0$, given $R > 0, Q^i \geq 0$, and a threshold $\varepsilon > 0, \alpha \in (0, 1]$

2. Kernel matrix evaluation and control policy update: solve H^{i+1} by

$$\begin{bmatrix} x_T(k) \\ u_T(k) \end{bmatrix}^T H^{i+1} \begin{bmatrix} x_T(k) \\ u_T(k) \end{bmatrix} - \begin{bmatrix} x_T(k+1) \\ u_T(k+1) \end{bmatrix}^T H^{i+1} \begin{bmatrix} x_T(k+1) \\ u_T(k+1) \end{bmatrix} = x_T(k)^T Q^i x_T(k) + u_T(k)^T R u_T(k).$$

and update u^{i+1} by $u^{i+1} = -K^{i+1}x = -(H_{uu}^{i+1})^{-1} H_{ux}^{i+1}x$

3. Cost function weight correction: compute $H_p^{i+1} = H^{i+1} - \begin{bmatrix} Q^i & 0 \\ 0 & R \end{bmatrix}$, and solve Q^{i+1} by

$$x(k)^T Q^{i+1} x(k) = (1 - \alpha) x(k)^T Q^i x(k) - \alpha u(k)^T R u(k) + \alpha \begin{bmatrix} x(k-1) \\ u(k-1) \end{bmatrix}^T H_p^{i+1} \begin{bmatrix} x(k-1) \\ u(k-1) \end{bmatrix} - \alpha \begin{bmatrix} x(k) \\ u(k) \end{bmatrix}^T H_p^{i+1} \begin{bmatrix} x(k) \\ u(k) \end{bmatrix}$$

4. Set $i = i + 1$, repeat 1-3 until $\|Q^{i+1} - Q^i\| \leq \varepsilon$.

Model-Free Inverse RL for DT State-Feedback Control Systems

■ Algorithm Properties

- ✓ **The data-driven algorithm uses only data, not system model/dynamics**
- ✓ **Each updated control policy stabilizes the controlled system**
- ✓ **Single-loop learning procedure reduces computation and improves learning efficiency**
- ✓ **Iteration is convergent**
- ✓ **The learned cost function may not be unique, but all of them are the desired solution even if the true cost function of the expert is non-identifiable**
- ✓ **Degenerate solution ($Q=0$, $R=0$) is avoided**

Model-Free Inverse RL for DT State-Feedback Control Systems

■ Simulation Experiment

The dynamics of the Expert and Learner systems are (F-16 aircraft system dynamics)

$$A = \begin{bmatrix} 0.906488 & 0.0816012 & -0.0005 \\ 0.0741349 & 0.90121 & -0.000708383 \\ 0 & 0 & 0.132655 \end{bmatrix}, B = \begin{bmatrix} -0.00150808 \\ -0.0096 \\ 0.867345 \end{bmatrix}, C = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0.6 & 0 \\ -1.26 & -0.9788 & 0.4852 \end{bmatrix}.$$

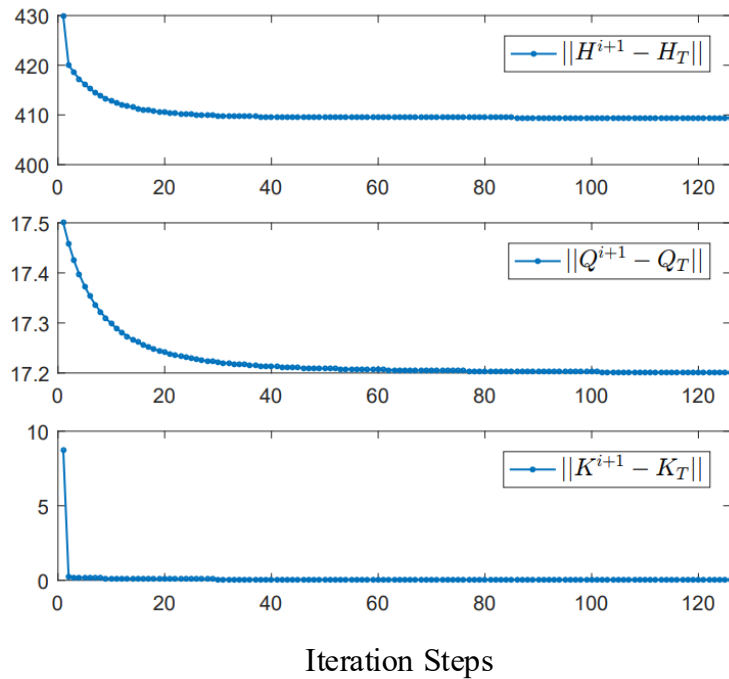
$$\text{Expert's input: } u_T(k) = -K_T x_T(k) = -[-0.1688 \quad -0.2009 \quad 0.1285] x_T(k) \quad \text{Expert's weights: } Q_T = \begin{bmatrix} 15 & 0 & 0 \\ 0 & 18 & 0 \\ 0 & 0 & 14 \end{bmatrix}, R_T = 2I_1$$

$$\text{Initial parameters for learning: } Q^0 = \begin{bmatrix} 0.1 & 0 & 0 \\ 0 & 0.5 & 0 \\ 0 & 0 & 1 \end{bmatrix}, R = I_1, \alpha=1 \quad \text{Threshold: } \varepsilon = 6 \times 10^{-4}$$

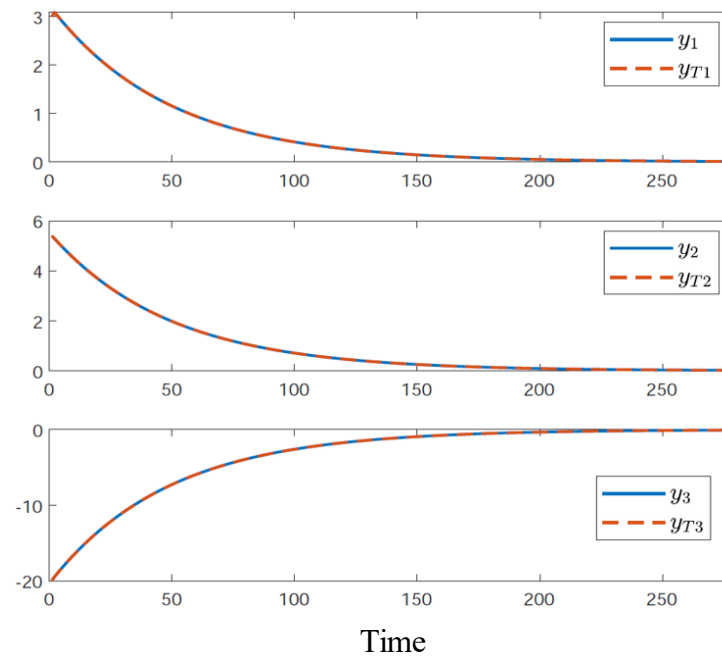
Simulation goal: Using expert's demonstrations, find controlled system a group of Q , R , K using the data-driven model-free IRL algorithm to realize trajectory imitation.

Model-Free Inverse RL for DT State-Feedback Control Systems

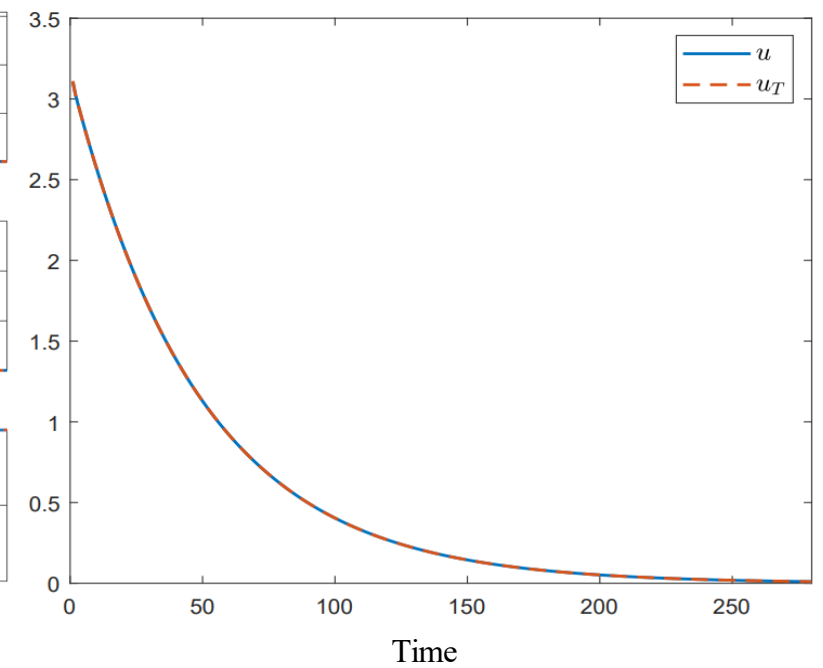
■ Simulation Experiment



Convergence of H^i, Q^i, K^i to H_T, Q_T, K_T



Imitation performance of output and input



The learned control gain $K = [-0.1671 \quad -0.1987 \quad 0.1168]$

Model-Free Inverse RL for DT State-Feedback Control Systems

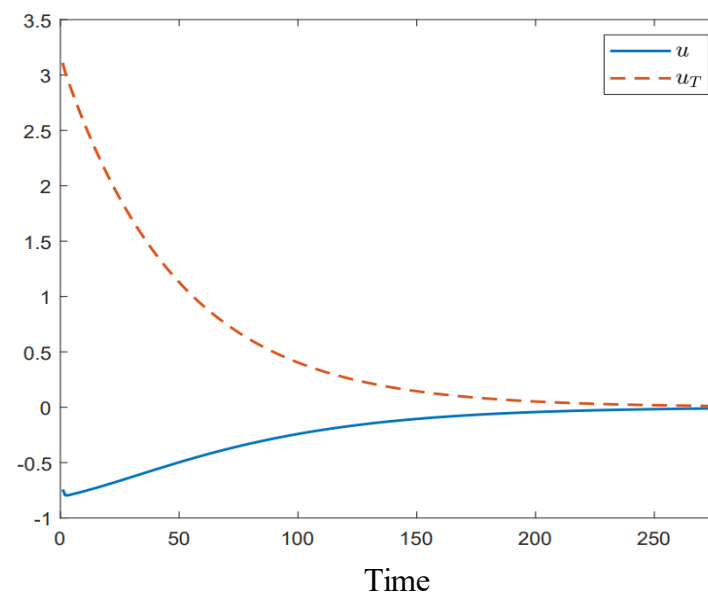
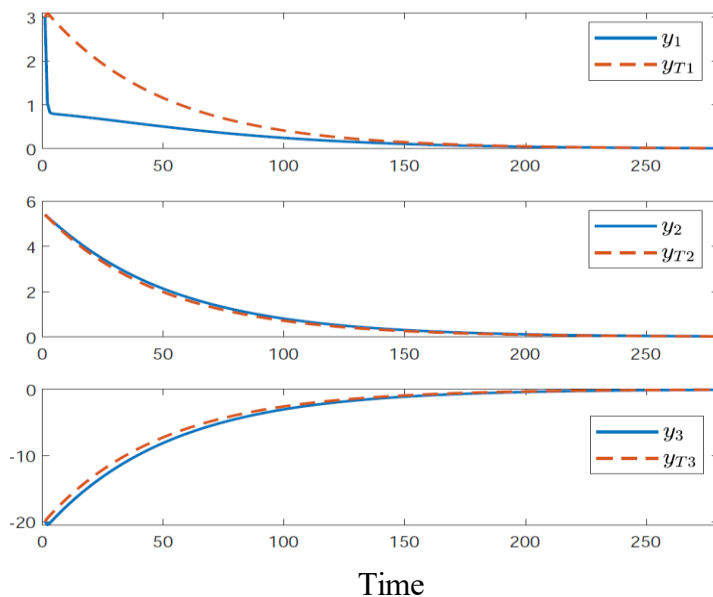
■ Simulation Experiment (for control performance improvement by correcting cost function)

Comparison method 1:

B. Kiumarsi, F. L. Lewis, H. Modares, A. Karimpour, and M. B. Naghibi-Sistani, "Reinforcement Q-learning for optimal tracking control of linear discrete-time systems with unknown dynamics," Automatica, vol. 50, no. 4, pp. 1167–1175, 2014.

Parameter setting:

$$Q = \begin{bmatrix} 0.1 & 0 & 0 \\ 0 & 0.5 & 0 \\ 0 & 0 & 1 \end{bmatrix}, R = I_1, \gamma = 0.9$$



Imitation performance of output and input

	Output RMSE	Input RMSE
Inverse RL	5.0476×10^{-5}	1.1473×10^{-4}
Com 1	0.4053	1.2422

$$\text{RMSE} = \frac{1}{n} \sum_{i=1}^n \sqrt{\frac{1}{a+1} \sum_{k=k^*}^{k^*+a} |y_i(k) - y_{Ti}(k)|^2}$$

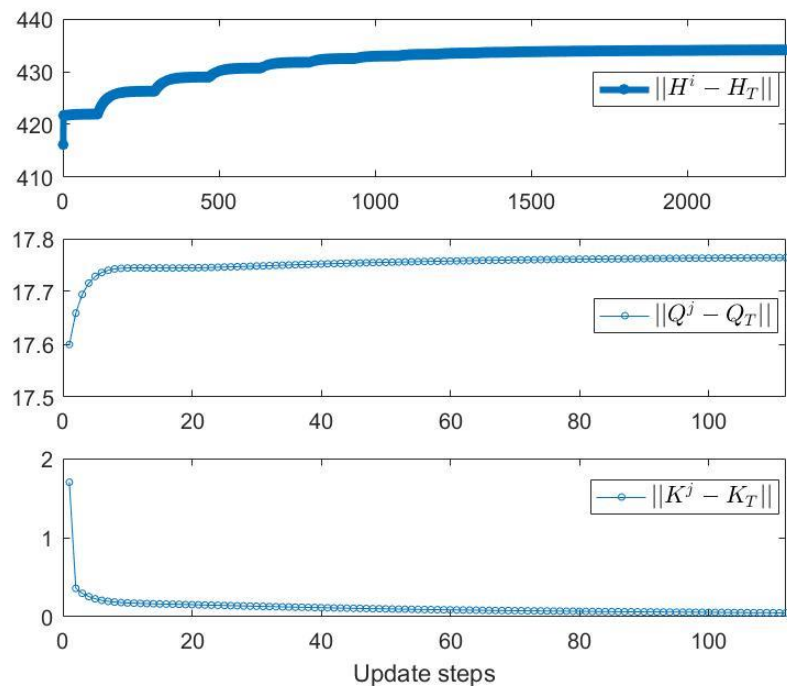
$$k^* = 1, a = 280$$

Model-Free Inverse RL for DT State-Feedback Control Systems

■ Simulation Experiment (for learning efficiency improvement by the single-loop learning structure)

Comparison method 2:

Apply the bilevel two-loop learning structure [1, 2] to the proposed formulas.



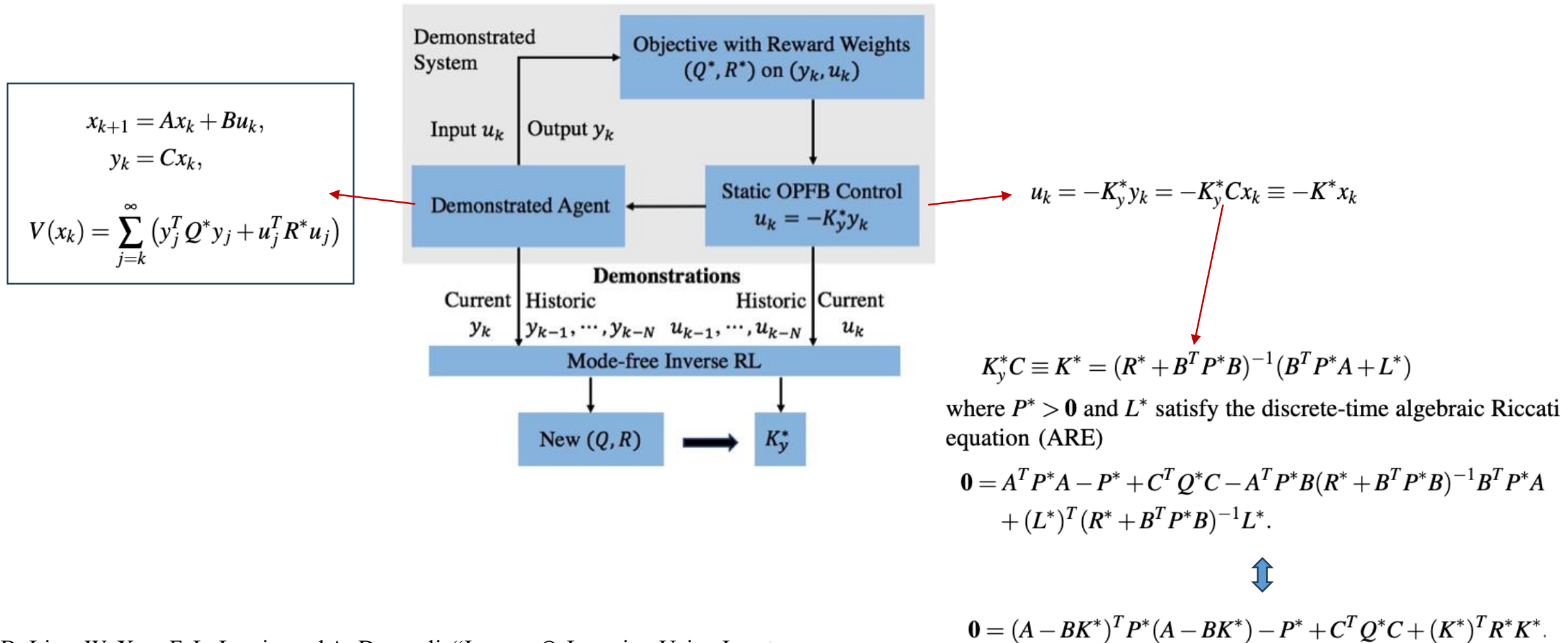
	Outer iterations j	Inner iterations i	Total	time
Inverse RL	\	\	81	10.360s
Com 2	112	2315	2427	190.576s

Under threshold $\|K^j - K_d\| \leq 0.05$ for each inner-loop converged K^j

[1] P. Abbeel and A. Y. Ng, “Apprenticeship Learning via Inverse Reinforcement Learning,” the 21st International Conference on Machine Learning, Banff, Canada, 2004.

[2] K. Mombaur, A. Truong, J P Laumond. From Human to Humanoid Locomotion—An Inverse Optimal Control Approach. Auton Robots, 2010, 28:369–383.

Model-Free Inverse RL for DT Output-Feedback Control Systems



Model-Free Inverse RL for DT Output-Feedback Control Systems

■ Algorithm Design

Goal: Using expert's demonstrations, find a group of Q , R_e such that

$$\begin{aligned} \mathbf{0} &= (A - BK^*)^T P_e (A - BK^*) - P_e + C^T Q_e C + (K^*)^T R_e K^*, \\ K_e C &= (R_e + B^T P_e B)^{-1} (B^T P_e A + L_e), \\ L_e &= (R_e + B^T P_e B) K_e C - B^T P_e A. \end{aligned}$$

Algorithm 1 Model-Based Inverse RL Algorithm for Discrete-Time Static OPFB Control Systems

1: **Initialization:** Select arbitrary $R_e = R_e^T \in \mathbb{R}^{m \times m} > \mathbf{0}$, a small threshold δ , and a tuning parameter $\alpha \in (0, 1]$. Give (A, B, C) and K^* . Set $i = 0$, $Q_0 = \mathbf{0}$, and $L_0 = \mathbf{0}$.

2: **Evaluate P_i using Lyapunov equation**

$$\mathbf{0} = (A - BK^*)^T P_i (A - BK^*) - P_i + C^T Q_i C + (K^*)^T R_e K^*.$$

3: **Improve K_{i+1} and L_{i+1} using static OPFB control**

$$\begin{aligned} K_{i+1} &= (R_e + B^T P_i B)^{-1} (B^T P_i A + L_i) C^T (C C^T)^{-1}, \\ L_{i+1} &= (R_e + B^T P_i B) K_{i+1} C - B^T P_i A. \end{aligned}$$

4: **Improve Q_{i+1} relying on inverse optimal control**

$$\begin{aligned} Q_{i+1} &= \alpha (C C^T)^{-1} C [P_i - (A - BK_{i+1} C)^T P_i (A - BK_{i+1} C) \\ &\quad - C^T K_{i+1}^T R_e K_{i+1} C - L_i^T K^* - (K^*)^T L_i + L_i^T K_{i+1} C \\ &\quad + C^T K_{i+1}^T L_i] C^T (C C^T)^{-1} + (1 - \alpha) Q_i. \end{aligned}$$

5: **Stop if $\|K_{i+1} - K^*\| \leq \delta$, or set $i \rightarrow i + 1$ and go to Step 2.**

Only measurement data of expert's u_k, y_k are available

Model-Free Inverse RL for DT Output-Feedback Control Systems

Replace

2: Evaluate P_i using Lyapunov equation

$$\mathbf{0} = (A - BK^*)^T P_i (A - BK^*) - P_i + C^T Q_i C + (K^*)^T R_e K^*.$$

Model-free version to update P^i ($\bar{H}^i$):

System State
Reconstruction

$$x_k = D_u U_{k-1:k-N} + D_y Y_{k-1:k-N}$$

$$U_{k-1:k-N} = \begin{bmatrix} u_{k-1} \\ u_{k-2} \\ \vdots \\ u_{k-N} \end{bmatrix}, \quad Y_{k-1:k-N} = \begin{bmatrix} y_{k-1} \\ y_{k-2} \\ \vdots \\ y_{k-N} \end{bmatrix}$$

$$D_u = V_N - A^N (M_N^T M_N)^{-1} M_N^T T_N, \\ D_y = A^N (M_N^T M_N)^{-1} M_N^T,$$

$$V_N = \begin{bmatrix} B & AB & \dots & A^{N-1}B \end{bmatrix}, \\ M_N = \begin{bmatrix} CA^{N-1} \\ CA^{N-2} \\ \vdots \\ CA \\ C \end{bmatrix}, \quad T_N = \begin{bmatrix} \mathbf{0} & CB & CAB & \dots & CA^{N-2}B \\ \mathbf{0} & \mathbf{0} & CB & \dots & CA^{N-3}B \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ \mathbf{0} & \dots & \dots & \mathbf{0} & CB \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix}$$

$$\begin{bmatrix} x_k \\ u_k \end{bmatrix}^T H^i \begin{bmatrix} x_k \\ u_k \end{bmatrix} = \begin{bmatrix} x_{k+1} \\ u_{k+1} \end{bmatrix}^T H^i \begin{bmatrix} x_{k+1} \\ u_{k+1} \end{bmatrix} + x_k^T C^T Q_i C x_k + u_k^T R_e u_k$$

$$z_k = \begin{bmatrix} U_{k-1:k-N}^T & Y_{k-1:k-N}^T & u_k^T \end{bmatrix}^T$$



$$z_k^T \bar{H}^i z_k - z_{k+1}^T \bar{H}^i z_{k+1} = y_k^T Q_i y_k + u_k^T R_e u_k.$$

$$H^i \equiv \begin{bmatrix} H_{xx}^i & H_{xu}^i \\ H_{ux}^i & H_{uu}^i \end{bmatrix} \in \mathbb{R}^{(n+m) \times (n+m)},$$

$$H_{xx}^i \equiv C^T Q_i C + A^T P_i A \in \mathbb{R}^{n \times n},$$

$$H_{xu}^i = (H_{ux}^i)^T \equiv A^T P_i B \in \mathbb{R}^{n \times m},$$

$$H_{uu}^i \equiv R_e + B^T P_i B \in \mathbb{R}^{m \times m}.$$

$$\bar{H}^i \equiv \begin{bmatrix} H_{UU}^i & H_{UY}^i & H_{Uu}^i \\ H_{YU}^i & H_{YY}^i & H_{Yu}^i \\ H_{uU}^i & H_{uY}^i & H_{uu}^i \end{bmatrix} \in \mathbb{R}^{(mN+pN+m) \times (mN+pN+m)},$$

$$H_{UU}^i \equiv D_u^T H_{xx}^i D_u \in \mathbb{R}^{mN \times mN}, \quad H_{UY}^i \equiv D_u^T H_{xx}^i D_y \in \mathbb{R}^{mN \times pN},$$

$$H_{Uu}^i \equiv D_u^T H_{xu}^i \in \mathbb{R}^{mN \times m}, \quad H_{YY}^i \equiv D_y^T H_{xx}^i D_y \in \mathbb{R}^{pN \times pN},$$

$$H_{Yu}^i \equiv D_y^T H_{xu}^i \in \mathbb{R}^{pN \times m}, \quad H_{YU}^i = (H_{UY}^i)^T \in \mathbb{R}^{pN \times mN},$$

$$H_{uU}^i = (H_{Uu}^i)^T \in \mathbb{R}^{m \times mN}, \quad H_{uY}^i = (H_{Yu}^i)^T \in \mathbb{R}^{m \times pN}.$$

Model-Free Inverse RL for DT Output-Feedback Control Systems

Algorithm 1 Model-Based Inverse RL Algorithm for Discrete-Time Static OPFB Control Systems

1: **Initialization:** Select arbitrary $R_e = R_e^T \in \mathbb{R}^{m \times m} > \mathbf{0}$, a small threshold δ , and a tuning parameter $\alpha \in (0, 1]$. Give (A, B, C) and K^* . Set $i = 0$, $Q_0 = \mathbf{0}$, and $L_0 = \mathbf{0}$.

2: **Evaluate P_i using Lyapunov equation**

$$\mathbf{0} = (A - BK^*)^T P_i (A - BK^*) - P_i + C^T Q_i C + (K^*)^T R_e K^*.$$

3: **Improve K_{i+1} and L_{i+1} using static OPFB control**

$$K_{i+1} = (R_e + B^T P_i B)^{-1} (B^T P_i A + L_i) C^T (C C^T)^{-1},$$

$$L_{i+1} = (R_e + B^T P_i B) K_{i+1} C - B^T P_i A.$$

4: **Improve Q_{i+1} relying on inverse optimal control**

$$\begin{aligned} Q_{i+1} = & \alpha (C C^T)^{-1} C [P_i - (A - BK_{i+1} C)^T P_i (A - BK_{i+1} C) \\ & - C^T K_{i+1}^T R_e K_{i+1} C - L_i^T K^* - (K^*)^T L_i + L_i^T K_{i+1} C \\ & + C^T K_{i+1}^T L_i] C^T (C C^T)^{-1} + (1 - \alpha) Q_i. \end{aligned}$$

5: **Stop if $\|K_{i+1} - K_y^*\| \leq \delta$, or set $i \rightarrow i + 1$ and go to Step 2.**

Model-free version to update K_{i+1} and L_{i+1} :

$$\begin{aligned} K_{i+1} C x_k &= K_{i+1} y_k \\ &= (R_e + B^T P_i B)^{-1} (B^T P_i A + L_i) x_k \\ &= (R_e + B^T P_i B)^{-1} (B^T P_i A D_u U_{k-1:k-N} \\ &\quad + B^T P_i A D_y Y_{k-1:k-N}) + (R_e + B^T P_i B)^{-1} L_i x_k \\ &= (H_{uu}^i)^{-1} (H_{uU}^i U_{k-1:k-N} + H_{uY}^i Y_{k-1:k-N}) \\ &\quad + (H_{uu}^i)^{-1} L_i (x_k), \end{aligned}$$

$$\begin{aligned} L_{i+1} (x_k) &= [(R_e + B^T P_i B) K_{i+1} C - B^T P_i A] x_k \\ &= (R_e + B^T P_i B) K_{i+1} C x_k - (B^T P_i A D_u U_{k-1:k-N} \\ &\quad + B^T P_i A D_y Y_{k-1:k-N}) \\ &= H_{uu}^i K_{i+1} y_k - (H_{uU}^i U_{k-1:k-N} + H_{uY}^i Y_{k-1:k-N}). \end{aligned}$$

Model-Free Inverse RL for DT Output-Feedback Control Systems

Algorithm 1 Model-Based Inverse RL Algorithm for Discrete-Time Static OPFB Control Systems

1: **Initialization:** Select arbitrary $R_e = R_e^T \in \mathbb{R}^{m \times m} > \mathbf{0}$, a small threshold δ , and a tuning parameter $\alpha \in (0, 1]$. Give (A, B, C) and K^* . Set $i = 0$, $Q_0 = \mathbf{0}$, and $L_0 = \mathbf{0}$.

2: **Evaluate P_i using Lyapunov equation**

$$\mathbf{0} = (A - BK^*)^T P_i (A - BK^*) - P_i + C^T Q_i C + (K^*)^T R_e K^*$$

3: **Improve K_{i+1} and L_{i+1} using static OPFB control**

$$K_{i+1} = (R_e + B^T P_i B)^{-1} (B^T P_i A + L_i) C^T (C C^T)^{-1},$$

$$L_{i+1} = (R_e + B^T P_i B) K_{i+1} C - B^T P_i A.$$

4: **Improve Q_{i+1} relying on inverse optimal control**

$$\begin{aligned} Q_{i+1} = & \alpha (C C^T)^{-1} C [P_i - (A - BK_{i+1} C)^T P_i (A - BK_{i+1} C) \\ & - C^T K_{i+1}^T R_e K_{i+1} C - L_i^T K^* - (K^*)^T L_i + L_i^T K_{i+1} C \\ & + C^T K_{i+1}^T L_i] C^T (C C^T)^{-1} + (1 - \alpha) Q_i. \end{aligned}$$

5: **Stop if $\|K_{i+1} - K_y^*\| \leq \delta$, or set $i \rightarrow i + 1$ and go to Step 2.**

Model-free version to update Q_{i+1} :

$$C^T Q_{i+1} C = \alpha (K_{i+1} C - K^*)^T (R_e + B^T P_i B) (K_{i+1} C - K^*) + C^T Q_i C.$$

Multiply x_k

$$\begin{aligned} y_k^T Q_{i+1} y_k &= \alpha (K_{i+1} C x_k + u_k)^T H_{uu}^i (K_{i+1} C x_k + u_k) + y_k^T Q_i y_k \\ &= \alpha (K_{i+1} y_k + u_k)^T H_{uu}^i (K_{i+1} y_k + u_k) + y_k^T Q_i y_k. \end{aligned}$$

Data-Driven Model-Free Inverse RL for Linear Systems

■ Results for Continuous-Time Linear Systems

For CT linear system with unknown dynamics:

$$\dot{x}(x) = Ax(t) + Bu(t)$$

Algorithm 1 Model-based Inverse Value Iteration Algorithm

Input: Any $R = R^T > 0$. Threshold $\varepsilon > 0$. (A, B) . K_d .

Output: Converged values Q^∞ and K^∞ .

1: **Initialization:** $i = 0$. $Q^0 \geq 0$.

2: **Evaluate P^i using Lyapunov Equation**

$$0 = (A - BK_d)^T P^i + P^i (A - BK_d) + Q^i + (K_d)^T R K_d.$$

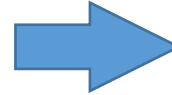
3: **Improve K^{i+1} using Optimal Control**

$$K^{i+1} = R^{-1} B^T P^i.$$

4: **Improve Q^{i+1} using IOC**

$$Q^{i+1} = -(A - BK^{i+1})^T P^i - P^i (A - BK^{i+1}) - (K^{i+1})^T R K^{i+1}.$$

5: **Stop if $\|Q^{i+1} - Q\| \leq \varepsilon$, or set $i \rightarrow i + 1$ and go to Step 2.**



Algorithm 2 Model-free Inverse Q -learning Algorithm

Input: Any $R = R^T > 0$. Threshold $\varepsilon > 0$, $z = [x_d^T, u_d^T]^T$

Output: Converged values Q^∞ and K^∞ .

1: **Initialization:** $i = 0$. $Q^0 \geq 0$.

2: **Evaluate H^i using Integral RL by**

$$\begin{aligned} & z^T(t) H^i z(t) - z^T(t + \Delta t) H^i z(t + \Delta t) \\ &= \int_t^{t+\Delta t} (x_d^T Q^i x_d + u_d^T R u_d) d\tau. \end{aligned}$$

3: **Evaluate u^{i+1} (i.e., K^{i+1}) using Optimal Control by**

$$u^{i+1} = -(H_{u_d u_d}^i)^{-1} (H_{x_d u_d}^i)^T x_d.$$

4: **Improve Q^{i+1} using IOC by**

$$\begin{aligned} & \int_t^{t+\Delta t} x_d^T Q^{i+1} x_d d\tau \\ &= \int_t^{t+\Delta t} x_d^T Q^i x_d d\tau + \int_t^{t+\Delta t} (u_d - u^{i+1})^T R (u_d - u^{i+1}) d\tau. \end{aligned}$$

5: **Stop if $\|Q^{i+1} - Q^i\| \leq \varepsilon$, or set $i \rightarrow i + 1$ and go to Step 2.**

Data-Driven Model-Free Inverse RL for Linear Systems

■ Results for Continuous-Time Linear Systems

For CT linear system with disturbances:

$$\dot{x}(x) = Ax(t) + Bu(t) + Dd(t)$$

Algorithm 1 Model-based IRL for inverse H_∞ control

- 1) Initialize with $R > 0$, $\gamma > 0$, $Q^0 > 0$, $L^0 = 0$, and $i = 0$.
- 2) **Policy correction:** Update policy P^i by

$$\begin{aligned} (A - BK_T)^T P^i + P^i (A - BK_T) \\ = -Q^i - K_T^T R K_T - \gamma^2 (L^i)^T L^i. \end{aligned}$$

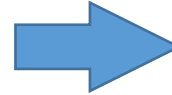
- 3) **Action update:** Update u^{i+1} , K^{i+1} and L^{i+1} by

$$\begin{aligned} u^{i+1} &= -K^{i+1}x = -R^{-1}B^T P^i x, \\ L^{i+1} &= \frac{1}{\gamma^2} D^T P^i. \end{aligned}$$

- 4) **Cost function weight construction:** Update Q^{i+1} by

$$Q^{i+1} = -A^T P^i - P^i A + (K^{i+1})^T R K^{i+1} - \gamma^2 (L^{i+1})^T L^{i+1}.$$

- 5) Stop if it converges. Otherwise, set $i = i + 1$ and repeat steps 2 to 4.
-



Algorithm 2 Model-free IRL algorithm for inverse H_∞ control

- 1: Initialize with $R > 0$, $\gamma > 0$, $Q^0 > 0$, and $L^0 = 0$, and set $i = 0$.
- 2: **Policy correction:** Update policy P^i , control gain K^{i+1} and disturbance gain L^{i+1} by

$$\begin{aligned} & x_T(t+T)^T P^i x_T(t+T) - x_T(t)^T P^i x_T(t) \\ & - 2 \int_t^{t+T} e^T R K^{i+1} x_T d\tau - 2\gamma^2 \int_t^{t+T} d_T^T L^{i+1} x_T d\tau \\ & = - \int_t^{t+T} (x_T^T Q^i x_T + (u_T - e)^T R (u_T - e) \\ & \quad + \gamma^2 x_T^T (L^i)^T L^i x_T) d\tau. \end{aligned}$$

- 3: **Cost function weight construction:** Update cost function weight Q^{i+1} by

$$\begin{aligned} & \int_t^{t+T} x^T Q^{i+1} x d\tau \\ & = -[x(t+T)^T P^i x(t+T) - x(t)^T P^i x(t) \\ & \quad - \int_t^{t+T} (2u^T R K^{i+1} x + x^T (K^{i+1})^T R K^{i+1} x) d\tau \\ & \quad - \gamma^2 \int_t^{t+T} (2d^T L^{i+1} x - x^T (L^{i+1})^T L^{i+1} x) d\tau]. \end{aligned}$$

- 4: Stop if it converges. Otherwise, set $i = i + 1$ and repeat steps 2 to 3.
-

Two-player Inverse RL

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Thank you!